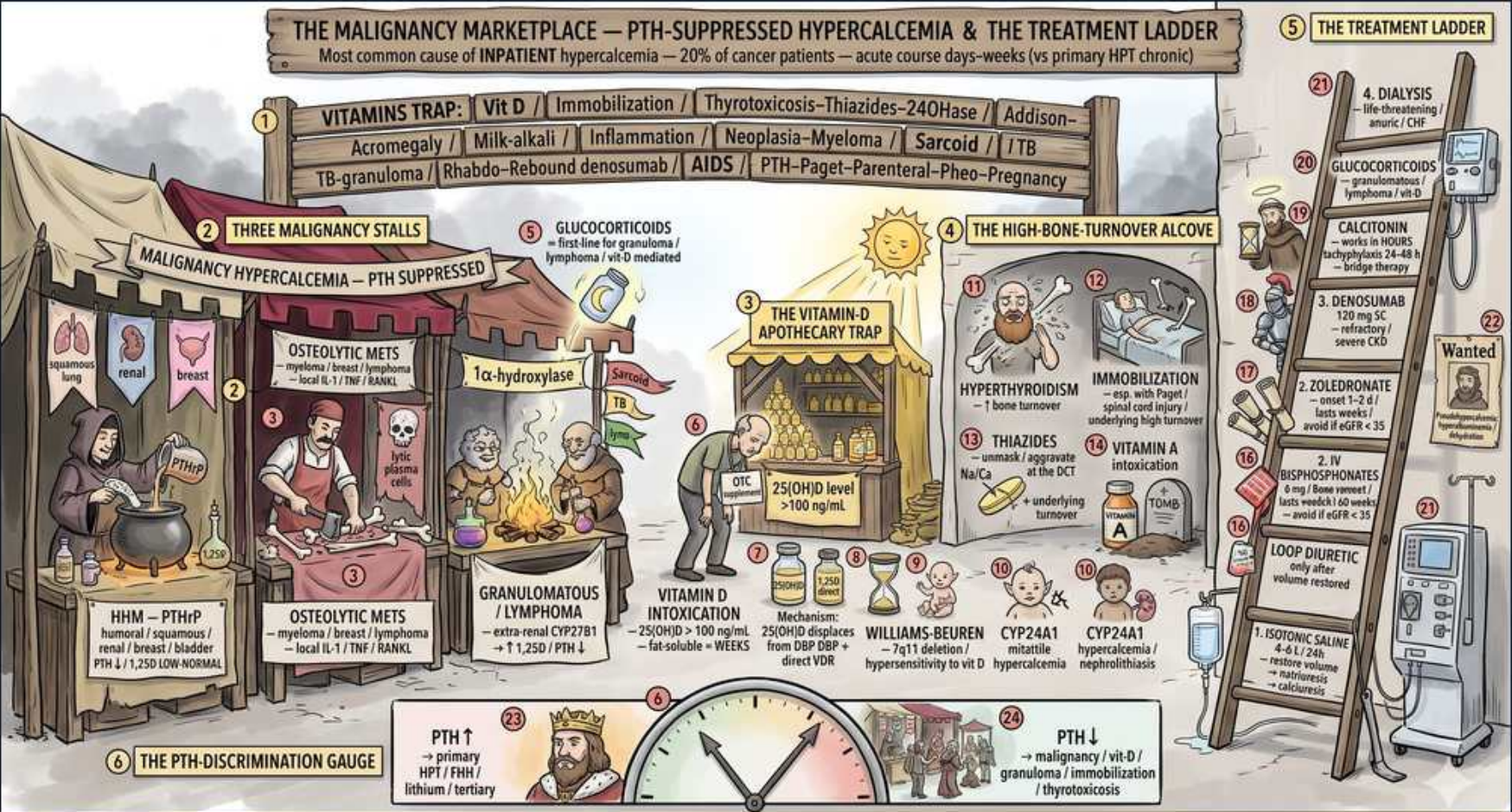


Parathyroid/Bone — Hypercalcemia of Malignancy (Marketplace)



Inpatient acute hypercalcemia → think malignancy. Granuloma / lymphoma → high 1,25D + LOW 25(OH)D + glucocorticoid-responsive. Vit D intox → high 25(OH)D. SALINE FIRST, then bisphosphonate. Calcitonin = bridge only.

1. Inpatient Ca cause

WHAT YOU SEE

The subtitle under THE MALIGNANCY MARKETPLACE reads 'Most common cause of INPATIENT hypercalcemia — 20% of cancer patients — acute course days-weeks vs primary HPT chronic.'

WHAT IT ENCODES

Malignancy is the #1 cause of INPATIENT hypercalcemia (primary hyperPTH is #1 in outpatients/asymptomatic). It develops over days-to-weeks, so Ca is often very high (>13-14 mg/dL) and patients are symptomatic — polyuria, dehydration, confusion, constipation ('stones, bones, groans, psychiatric overtones'). Up to ~20-30% of cancer patients develop it, and it portends poor prognosis. Mnemonic for symptom severity: Ca >14 mg/dL or rapid rise = emergency.

BOARD TRAP

WRONG: calling rapid, severe, symptomatic hypercalcemia 'primary hyperparathyroidism.' Primary hyperPTH is typically chronic, mild (Ca 10.5-11.5), asymptomatic, found incidentally — discriminator is the TEMPO and SEVERITY plus a suppressed PTH. A sick, dehydrated patient with Ca 14 and confusion is malignancy until proven otherwise.

2. PTH suppressed gauge

WHAT YOU SEE

THE PTH-DISCRIMINATION GAUGE at the bottom — a dial split so HIGH PTH points to primary/tertiary hyperPTH and LOW PTH points to malignancy / vitamin D / granulomatous — the needle sorts the whole differential.

WHAT IT ENCODES

First branch point in any hypercalcemia workup is the intact PTH. PTH HIGH/inappropriately-normal = PTH-dependent (primary hyperPTH, tertiary hyperPTH, lithium, FHH). PTH LOW/suppressed (<20 pg/mL) = PTH-independent: malignancy, vitamin D excess, granulomatous disease, thyrotoxicosis, immobilization, milk-alkali, vitamin A. In malignancy the high Ca feeds back on the normal parathyroids and shuts them off.

BOARD TRAP

WRONG: 'low PTH with high Ca means primary hyperparathyroidism.' By definition primary hyperPTH has an inappropriately HIGH or non-suppressed PTH. A suppressed PTH points you AWAY from the gland and toward malignancy/vitamin D — order PTHrP and 25/1,25-vitamin D next, not a parathyroid sestamibi scan.

3. HHM = PTHrP

WHAT YOU SEE

The first market stall on the left, tagged HHM — PTHrP — humoral hypercalcemia, squamous / renal / breast, low 1,25-OH-D, low-normal PTH — the PTHrP-mimics-PTH stall.

WHAT IT ENCODES

Humoral hypercalcemia of malignancy (HHM) is the MOST COMMON mechanism (~80%). Tumor secretes PTHrP, which binds the PTH1 receptor and mimics PTH: raises bone resorption and renal Ca reabsorption, and causes phosphaturia (low phosphate). Classic tumors: SQUAMOUS cell carcinoma (lung, head & neck, esophagus, cervix), RENAL cell carcinoma, bladder, and some breast cancers. Labs: high Ca, low PO4, LOW intact PTH, elevated PTHrP, and characteristically LOW 1,25-OH-D (PTHrP does not stimulate 1-alpha-hydroxylase well).

BOARD TRAP

WRONG: relying on the standard intact-PTH assay to detect PTHrP. The routine PTH assay does NOT cross-react with PTHrP, so PTH reads LOW in HHM — you must order a SEPARATE PTHrP level. Also wrong to expect high 1,25-vitamin D in HHM (that's the granulomatous/lymphoma mechanism); HHM has low 1,25-D.

4. Osteolytic mets

WHAT YOU SEE

The next stall showing bones being chipped apart, tagged OSTEOLYTIC METS — myeloma / breast — local IL-1, IL-6, TNF, RANKL — tumor in the marrow dissolving bone directly.

WHAT IT ENCODES

Local osteolytic hypercalcemia (~20%): tumor cells in bone secrete local cytokines (IL-1, IL-6, TNF, and RANKL) that activate osteoclasts and dissolve bone, dumping Ca into blood. Classic: breast cancer bone mets and MULTIPLE MYELOMA (myeloma cells make RANKL/MIP-1-alpha and suppress osteoblasts via DKK1 -> pure lytic 'punched-out' lesions). Both PTH and PTHrP are low here; the driver is local marrow biology, not a circulating hormone.

BOARD TRAP

WRONG: attributing myeloma hypercalcemia to PTHrP. Myeloma works by LOCAL osteoclast activation (RANKL/cytokines), so PTHrP is normal/low — clues are anemia, renal failure, high total protein/globulin gap, rouleaux, M-spike on SPEP/UPEP. Also note bone scan can be FALSELY negative in myeloma (lytic, not blastic) — use skeletal survey/MRI.

5. 1,25-D in lymphoma

WHAT YOU SEE

THE VITAMIN-D APOTHECARY-TRAP stall, tagged GRANULOMATOUS / LYMPHOMA — extrarenal 1α-hydroxylase (CYP27B1) → high 1,25(OH)2D — the rogue apothecary brewing active vitamin D unchecked.

WHAT IT ENCODES

Some lymphomas (especially Hodgkin and some non-Hodgkin) and granulomatous disease (sarcoidosis, TB, fungal) express extrarenal 1-alpha-hydroxylase (CYP27B1) in macrophages, converting 25-D to active 1,25-D unchecked by normal feedback. This raises gut Ca absorption -> hypercalcemia. Labs: high Ca, HIGH 1,25-OH-D, LOW PTH, often high/normal phosphate. Sarcoid hint: high ACE, hypercalciuria even more than hypercalcemia, bilateral hilar adenopathy.

BOARD TRAP

WRONG: treating the 1,25-D mechanism with the same algorithm as bone mets. This subtype is uniquely GLUCOCORTICOID-RESPONSIVE — prednisone shuts off macrophage 1-alpha-hydroxylase and Ca falls in days. Bisphosphonates work too but steroids are the targeted move. Do NOT expect a response to steroids in PTHrP (HHM) or osteolytic-met hypercalcemia.

6. Vitamins trap (causes)

WHAT YOU SEE

The VITAMINS TRAP banner of non-malignant causes across the top: Vit D, Milk-alkali, Immobilization, Thyrotoxicosis, Neoplasia, Myeloma, Sarcoid, Addison, Thiazides, granuloma — the reversible mimics to rule out.

WHAT IT ENCODES

Non-malignant PTH-independent (low-PTH) hypercalcemia mnemonic — 'VITAMINS TRAP': Vitamin D/A excess, Immobilization, Thyrotoxicosis, Addison's (adrenal insufficiency), Milk-alkali syndrome (high Ca intake + alkalosis + renal failure, classically calcium carbonate), Inflammatory/granulomatous (sarcoid, TB), Neoplasia, Sarcoid, and drugs like Thiazides. Milk-alkali is now the 3rd most common cause of hypercalcemia (calcium carbonate antacids/supplements).

BOARD TRAP

WRONG: anchoring on cancer the moment PTH comes back low. Many of these are reversible: stop the thiazide, stop vitamin D/A or calcium-carbonate antacids (milk-alkali), mobilize the patient, treat the thyrotoxicosis or Addison's. A careful med/supplement history can spare an unnecessary malignancy workup.

7. Saline first

WHAT YOU SEE

THE TREATMENT LADDER rung labelled SALINE FIRST — aggressive IV normal saline for volume repletion — the bottom, first-step rung of the ladder.

WHAT IT ENCODES

First-line acute therapy is aggressive IV isotonic SALINE (normal saline), typically 200-300 mL/hr to maintain urine output ~100-150 mL/hr. Hypercalcemia causes nephrogenic DI and osmotic diuresis -> profound volume depletion; rehydration restores GFR and dramatically increases urinary Ca excretion (saline natriuresis drags Ca with Na). Saline lowers Ca by ~1-3 mg/dL.

BOARD TRAP

WRONG: reaching for a loop diuretic ('furosemide') as first-line to 'flush out calcium.' Routine loop diuretics are NOT recommended — they only help AFTER full volume repletion or in volume-overloaded/CHF/renal patients, and given to a dehydrated patient they worsen the picture. Also wrong to start with a bisphosphonate alone (too slow). SALINE first.

8. Calcitonin bridge

WHAT YOU SEE

A ladder rung with a 'WANTED' poster for CALCITONIN — fast onset but short-lived (tachyphylaxis) — drawn as the quick but quickly-escaping bridge agent.

WHAT IT ENCODES

Salmon calcitonin (4 units/kg SC/IM q12h) is the FASTEST agent — lowers Ca within 4-6 hours by inhibiting osteoclasts and increasing renal Ca excretion. But the effect is small (~1-2 mg/dL) and short-lived: TACHYPHYLAXIS develops within ~48 hours. It buys time as a BRIDGE while the IV bisphosphonate (slow onset) takes effect.

BOARD TRAP

WRONG: relying on calcitonin as the durable/definitive therapy. Because of tachyphylaxis it loses efficacy in 1-2 days — it must be paired with saline plus a bisphosphonate or denosumab for sustained control. Treat it as a short bridge, never monotherapy for ongoing hypercalcemia.

9. Bisphosphonate (durable)

WHAT YOU SEE

A ladder rung tagged ZOLEDRONATE / pamidronate — takes 2-4 days, lasts weeks; renal caution — the slow-but-durable mainstay rung.

WHAT IT ENCODES

IV bisphosphonates are the DURABLE mainstay: ZOLEDRONIC ACID 4 mg IV over 15 min (preferred, more potent) or pamidronate 60-90 mg IV over 2-4 hr. They poison osteoclasts (inhibit farnesyl-pyrophosphate synthase in the mevalonate pathway). Onset is SLOW (2-4 days), nadir at ~4-7 days, but effect lasts 2-4 weeks. Adjust/avoid in renal impairment (GFR < ~30-35), pre-hydrate, and watch for osteonecrosis of the jaw and acute-phase flu-like reaction.

BOARD TRAP

WRONG: expecting a bisphosphonate to fix acute hypercalcemic crisis on its own. Because onset is 2-4 days it CANNOT be the sole acute agent — combine with saline (immediate) and calcitonin (fast bridge). Also wrong to give full-dose zoledronate in significant renal failure: dose-reduce or switch to denosumab.

10. Denosumab

WHAT YOU SEE

A ladder rung tagged DENOSUMAB — anti-RANKL — used when bisphosphonate-refractory or in renal failure (not renally cleared).

WHAT IT ENCODES

Denosumab is a monoclonal anti-RANKL antibody that blocks osteoclast formation/activation. Dose for malignant hypercalcemia is 120 mg SC. Key advantages: NOT renally cleared (safe in renal failure where bisphosphonates are limited) and effective in BISPHOSPHONATE-REFRACTORY hypercalcemia. Watch for hypocalcemia (especially with low vitamin D) and ONJ.

BOARD TRAP

WRONG: avoiding any antiresorptive because 'the kidneys are bad.' Denosumab is precisely the agent to use when GFR is too low for zoledronate or when bisphosphonates have failed. The flip side: it can cause profound, prolonged HYPOCALCEMIA — ensure vitamin D repletion and monitor Ca closely after dosing.

11. Glucocorticoids

WHAT YOU SEE

A ladder rung tagged GLUCOCORTICOIDS — for vitamin-D / lymphoma / 1,25-D-mediated hypercalcemia — the steroid rung that only works on the 1,25-D pathway.

WHAT IT ENCODES

Glucocorticoids (e.g., prednisone 20-40 mg/day or hydrocortisone) lower Ca specifically when the mechanism is excess 1,25-D: lymphoma, granulomatous disease (sarcoid, TB), and vitamin D intoxication. They inhibit macrophage 1-alpha-hydroxylase and decrease gut Ca absorption; onset 2-5 days.

BOARD TRAP

WRONG: giving steroids for PTHrP-mediated (HHM) or osteolytic-met hypercalcemia and expecting a response — they DON'T work for those. Glucocorticoids only target the 1,25-D/granulomatous pathway. Matching mechanism to therapy is the board point: high 1,25-D -> steroids; bone mets/PTHrP -> antiresorptives.

12. Dialysis

WHAT YOU SEE

The top ladder rung tagged DIALYSIS — life-threatening Ca with renal / heart failure — the last-resort rung using low-calcium dialysate.

WHAT IT ENCODES

Hemodialysis with a LOW-calcium (or calcium-free) dialysate is reserved for life-threatening hypercalcemia (Ca >18 mg/dL, severe neuro/cardiac symptoms) OR when aggressive saline is unsafe — renal failure or heart failure where you cannot volume-load. It rapidly removes calcium directly from blood.

BOARD TRAP

WRONG: defaulting to dialysis for ordinary hypercalcemia. It is a last-resort/rescue for refractory, dangerous hypercalcemia or when fluids are contraindicated (CHF, olig-anuric renal failure). For most patients, saline + calcitonin + bisphosphonate/denosumab is sufficient.

13. High-turnover causes

WHAT YOU SEE

THE HIGH-BONE-TURNOVER ALCOVE showing HYPERTHYROIDISM, IMMOBILIZATION, THIAZIDE, and VITAMIN A INTOXICATION — the reversible bone-turnover / renal-handling causes grouped together.

WHAT IT ENCODES

PTH-independent hypercalcemia from increased bone turnover or renal handling: THYROTOXICOSIS (thyroid hormone directly stimulates osteoclasts), IMMOBILIZATION (especially young patients with high baseline turnover, e.g., Paget's, or prolonged bed rest — bone resorption unopposed), THIAZIDES (decrease urinary Ca excretion at the DCT, unmask underlying hypercalcemia), and VITAMIN A intoxication (stimulates osteoclastic resorption).

BOARD TRAP

WRONG: missing the drug/endocrine cause. Always check TSH (thyrotoxicosis), review the med list for thiazides and vitamin A/retinoids, and ask about immobilization. These are reversible — stopping the thiazide or treating the hyperthyroidism corrects the Ca, sparing a malignancy hunt.
